

# Xiangru Jian

David R. Cheriton School of Computer Science  
University of Waterloo  
Waterloo, Canada

E-mail: xiangru.jian@uwaterloo.ca  
Website: Xiangru Jian  
Google Scholar: Xiangru Jian

## EDUCATION

- **University of Waterloo, Ph.D. in Computer Science.** Waterloo, Canada  
*Supervised by Prof. Tamer Özsu. Focus on VLM/LLM agents and multimodal data management. Sept. 2022 – Present*
- **City University of Hong Kong, M.Sc in Data Science.** Hong Kong SAR, China  
*Supervised by Prof. Yu Yang. Focus on data mining and graph learning. Aug. 2020 – Sept. 2021*
- **Tongji University, B.Eng.** Shanghai, China  
*Major in Civil Engineering and minor in Mathematics. Sept. 2014 – July. 2019*

## WORK EXPERIENCE

- **ServiceNow Research, Visiting Researcher (Internship)** Montreal, Canada  
*Focus on multimodal learning and GUI understanding. April. 2024 – May. 2025*
- **Hong Kong Institute of Data Science at CityU, Research Assistant** Hong Kong SAR, China  
*Focus on order prediction for industrial partners, such as Alibaba and HKTV Mall. July. 2021 – June. 2022*
- **City University of Hong Kong, Research Assistant** Hong Kong SAR, China  
*Focus on the landscape of DNNs. June. 2021 – Aug. 2021*

## RESEARCH AREAS

- **LLM and Multimodality:** Multimodal Reasoning; GUI Agents; Document Understanding; Natural Language Interface for Databases; Multi-agent System; Cross-modal Retrieval.
- **Machine Learning:** Graph Learning; Optimization; Set Learning and Generation.

## SELECTED PUBLICATIONS (\*CO-FIRST AUTHOR, ✉CORRESPONDING AUTHOR, †PROJECT LEAD)

- Data Management Agent **Xiangru Jian**, Zhengyuan Dong, M. Tamer Özsu. InteracSPARQL: an Interactive System for SPARQL Query Refinement Using Natural Language Explanations. **ACL 2026 Findings**. ([Paper Link](#), [Project Site](#))
- Agent Multimodal ✉**Xiangru Jian\***, Shravan Nayak\*, Kevin Qinghong Lin, Aarash Feizi, Kaixin Li, Patrice Bechard, Spandana Gella, Sai Rajeswar. CUA-Suite: Expert Trajectories and Pixel-Precise Grounding for Computer-Use Agents. **ICLR 2026 Workshop on Lifelong Agents**. ([Paper Link](#), [Project Site](#))
- VLM/LLM Graph Hao Xu\*, ✉**Xiangru Jian\*†**, et al. GraphOmni: A Comprehensive and Extendable Benchmark Framework for Large Language Models on Graph-theoretic Tasks. (**ICLR 2026**, [Paper Link](#), [Project Site](#))
- Agent Multimodal Aarash Feizi, Shravan Nayak, **Xiangru Jian**, et al. Grounding Computer Use Agents on Human Demonstrations. (Accepted by **ICLR 2026**, [Paper Link](#), [Project Site](#))
- Data Management Multimodal ✉**Xiangru Jian\*†**, Wei Pang\*, Zhengyuan Dong\*, Chao Zhang\*, M. Tamer Özsu. LazyVLM: Neuro-Symbolic Approach to Video Analytics. Accepted by **ICDE 2026**. ([Paper Link](#))
- Graph **Xiangru Jian\***, Xinjian Zhao\*, Wei Pang\*, Chaolong Ying, Yimu Wang, Yaoyao Xu, Tianshu Yu. Rethinking Spectral Augmentation for Contrast-based Graph Self-Supervised Learning. **TMLR**, 2025. ([Paper Link](#))
- Agent Multimodal ✉Shravan Nayak\*, ✉**Xiangru Jian\*** et al. UI-Vision: A Desktop-centric GUI Benchmark for Visual Perception and Interaction. **ICML 2025**. ([Paper Link](#), [Project Site](#))

- **Agent** **Multimodal** Wei Pang\*, Kevin Qinghong Lin\*, **Xiangru Jian\***, Xi He, Philip Torr. Paper2Poster: Towards Multimodal Poster Automation from Scientific Papers. Accepted by **NeurIPS** 2025. **Oral** at MAS Workshop at **ICML** 2025. ([Paper Link](#), [Project Site](#), 2.8k stars on [GitHub](#))
- **VLM/LLM** **Multimodal** **Xiangru Jian\***<sup>†</sup>, ... , Yoshua Bengio, ... , et al. BigDocs: An Open Dataset for Training Multimodal Models on Document and Code Tasks. **ICLR**, 2025. ([Paper Link](#), [Project Site](#))
- **Graph** **Multimodal** Xinjian Zhao\*, Wei Pang\*, Zhongkai Xue\*, **Xiangru Jian\***, et al. The Underappreciated Power of Vision Models for Graph Structural Understanding. **NeurIPS** 2025. ([Paper Link](#))
- **Retrieval** **Multimodal** **Graph** **Xiangru Jian** and Yimu Wang. InvGC: Robust Cross-Modal Retrieval by Inverse Graph Convolution. Findings of **EMNLP** 2023. ([Paper Link](#))
- **Retrieval** **Multimodal** Yimu Wang, **Xiangru Jian** and Xue Bo. Balance Act: Mitigating Hubness in Cross-Modal Retrieval with Query and Gallery Banks. **EMNLP** 2023. (**Oral**, [Paper Link](#))
- **Optimization** Keke Wu, **Xiangru Jian**, Rui Du, Jingrun Chen and Xiang Zhou. Understanding Loss Landscapes of Neural Network Models in Solving Partial Differential Equations. **IEEE BigData** 2023. (**Oral**, [Paper Link](#))
- **Optimization** Qixin Zhang, Zengde Deng, **Xiangru Jian**, Zaiyi Chen, Haoyuan Hu and Yu Yang. Communication-Efficient Decentralized Online Continuous DR-Submodular Maximization. **CIKM** 2023. ([Paper Link](#))
- **AI Application** **Graph** Lyuyi Zhu, Qixin Zhang, **Xiangru Jian**, Yu Yang. Graph convolutional network for traffic incidents duration classification. Engineering Applications of Artificial Intelligence. ([Paper Link](#))
- **AI Application** **Graph** Lyuyi Zhu, Qixin Zhang, **Xiangru Jian**, Yu Yang, Lishuai Li. Spatio-temporal Traffic Accidents Detection via Graph-based Generative Adversarial Network. Engineering Applications of Artificial Intelligence, Volumn 165.([Paper Link](#))

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#### SELECTED PREPRINTS (\*CO-FIRST AUTHOR, ✉CORRESPONDING AUTHOR, <sup>†</sup>PROJECT LEAD)

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- **AI Application** **Multimodal** **Xiangru Jian\***, Hao Xu\*, Wei Pang\*, Xinjian Zhao, Chengyu Tao, Qixin Zhang, Xikun Zhang, Chao Zhang, Guanzhi Deng, Alex Xue, Juan Du, Tianshu Yu, Garth Tarr, Qiuzhuang Sun, Dacheng Tao. FORGE: Fine-grained Multimodal Evaluation for Manufacturing Scenarios. (**Preprint**, [Paper Link](#), [Project Site](#))
- **Data Management** **Optimization** Yu Yin\*, **Xiangru Jian\***, Haoxiang Liu, Wei Gong, Yu Yang. A Pattern-based Subset Choice Model. (**Preprint**, Under review at [ESWA](#))
- **Graph** **Multimodal** Xinjian Zhao, Wei Pang, Zhixuan Yu, **Xiangru Jian**, et al. When Vision Meets Graphs: A Survey on Graph Reasoning and Learning.(**Preprint**, Under review at IJCAI 2026 Survey Track, [Paper Link](#))
- **VLM/LLM** **Multimodal** Tianyu Zhang, Suyuchen Wang, Chao Wang, Juan Rodriguez, Ahmed Masry, **Xiangru Jian**, Yoshua Bengio, Perouz Taslakian. SCOPE: Selective Cross-modal Orchestration of Visual Perception Experts. (**Preprint**, [Paper Link](#))

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#### SELECTED RESEARCH PROJECTS

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- **Research Intern at ServiceNow Research** Montreal, Canada  
*GUI Agents and Multimodal Reasoning* April 2024 – May 2025
  - **UI-Vision: A Desktop-centric GUI Benchmark for Visual Perception and Interaction**  
 Developed an open-source benchmark for evaluating autonomous GUI agents in desktop environments. Designed and curated a richly annotated dataset spanning 83 software applications to assess visual perception, layout grounding, and action prediction. (ICML 2025, [Paper Link](#), [Project Site](#))

- **GroundCUA: A High-Quality Dataset for Grounding Computer-Use Agents**  
Worked on the development of a large-scale instruction-tuning dataset for desktop UI localization, comprising 3M+ expert annotations across 80+ desktop applications. Trained state-of-the-art grounding models (GroundNext-3B and GroundNext-7B) using a two-stage recipe combining SFT and lightweight RL, achieving top performance on five grounding benchmarks, including OSWorld-G, UI-Vision, and ScreenSpot-Pro. (ICLR 2026, [Paper](#), [Project Site](#))
- **BigDocs: An Open Dataset for Training Multi-Modal Models on Document and Code Tasks**  
Co-led the effort of BigDocs—a 7.5-million sample, license-permissive dataset for multimodal document understanding. The dataset supports tasks such as document parsing and long-format code generation (e.g., Screenshot2HTML and Table2LaTeX), advancing research in structured output and document reasoning. (ICLR 2025, [Paper Link](#), [Project Site](#))

• **Graduate Research Assistant at University of Waterloo**

Waterloo, Canada

*Multimodal Data Management, Cross-Modal Retrieval and Graph Learning*

*Sept 2022 – Present*

- **Paper2Poster: Towards Multimodal Poster Automation from Scientific Papers**  
Proposed and developed Paper2Poster, the first benchmark and metric suite for academic poster generation, and PosterAgent, a visual-in-the-loop multi-agent pipeline. The system transforms research papers into editable posters, outperforms GPT-4o, and was accepted for Oral at the MAS Workshop ICML 2025 and later NeurIPS 2025. (NeurIPS 2025, [Paper Link](#), [Project Site](#)).
- **GraphOmni: Benchmarking LLMs on Graph-Theoretic Reasoning**  
Developed GraphOmni, a comprehensive and extensible evaluation framework that systematically benchmarks LLMs on graph reasoning tasks by varying graph types, serialization formats, and prompt schemes. Introduced an RL-based selector to dynamically choose optimal serialization-prompt combinations, yielding significant accuracy gains. (ICLR 2026, [Paper Link](#), [Project Site](#))
- **Robust Cross-Modal Retrieval**
  1. Formulated the representation degeneration problem in cross-modal retrieval and proposed InvGC, the first post-processing solution. (EMNLP 2023, [Paper Link](#))
  2. Developed DBNORM, a unified framework mitigating hubness by reducing similarity between hubs and queries. (EMNLP 2023, [Paper Link](#))
  3. Further proposed DREAM, a relevance-based augmentation framework using language and visual generative models, significantly improving video-text retrieval. (NAACL 2025, [Paper Link](#))
- **Graph Self-Supervised Learning**  
Conducted empirical studies revealing that simple edge perturbations (edge dropping and adding) offer comparable or superior performance to complex spectral augmentations in graph self-supervised learning, significantly reducing computational costs. Provided theoretical analysis supporting these findings and challenging conventional assumptions in spectral augmentation methods. (TMLR 2025, [Paper Link](#))
- **Application of Foundation Models on Data Management** (Supervised by Prof. Tamer Özsu)  
Proposed a datalog-like logic form that unifies various graph query languages (e.g., SPARQL) as an intermediate language for natural language interfaces to graph databases. Designed a pipeline for open-vocabulary video query across multi-frame systems, leveraging scene graph generation and a streaming graph OLAP architecture. ([ACL 2026 Findings](#) and [ICDE 2026 Demo](#))

• **Graduate Research Assistant at City University of Hong Kong**

Hong Kong SAR, China

*Data Mining and Optimization in E-commerce and Deep Learning*

*Sept 2020 – Sept 2022*

- **Submodular Maximization** (Supervised by Prof. Yu Yang)  
Developed one-shot decentralized algorithms (Mono-DMFW and DOBGA) that reduce per-round communications and gradient evaluations from  $T^{3/2}$  to 1, achieving  $(1 - 1/e)$ -approximation with  $O(T^{4/5})$  and  $O(\sqrt{T})$ -regret bounds, respectively. (CIKM 2023, [Paper Link](#))

PROFESSIONAL SERVICES

- Reviewer for: ICLR; ICML; NeurIPS; ECCV; ACL ARR; TMLR, TKDD; TBD; ACI Material Science Journal.